BEING UNIVERSALLY INJECTIVE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's alternative proof and the closing hint '(Hint: use that the base change of the diagonal is the diagonal of the base change.)' are part of the delivered proof and are quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The characterisation of universal injectivity by the diagonal, and the mapping property of points of a scheme with field-valued test objects. Morphisms, Lemma morphisms-lemma-universally-injective and Schemes, Section schemes-section-points, invoked at proof lines 5896-5925. Gives both the hypothesis to apply to X' -> S' and the passage between a = b and a' = b'.
- Every morphism to S from a field-valued point lifts after a field extension along a surjective morphism S' -> S. Schemes, Section schemes-section-points, invoked at proof lines 5898-5902. Produces the field extension K'/K and the morphism Spec(K') -> S' used to build the diagram.
- The general principle for a property that is fpqc local on the base. Same chapter, Lemma lemma-descending-properties-morphisms, invoked at proof lines 5920 and 5927-5928. Turns the verified special case into the locality statement, both in the main and in the alternative proof.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KV: sources/stacks-descent/pinned-original-source.tex, source lines 5826–5826. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

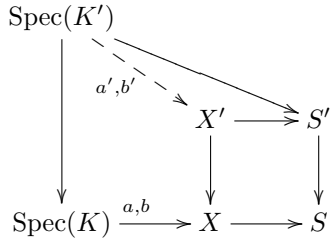
Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 01J5: sources/stacks-chow/tags.txt, source lines 2002–2002. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is universally injective" is fpqc local on the base.*

Proof. A base change of a universally injective morphism is universally injective (this is formal). Being universally injective is Zariski local on the base; this is clear from the definition. Finally, let $S' \to S$ be a flat surjective morphism of affine schemes, and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is universally injective. Let K be a field, and let $a, b : \mathrm{Spec}(K) \to X$ be two morphisms such that $f \circ a = f \circ b$. As $S' \to S$ is surjective and by the discussion in Schemes, Section 01J5 there exists a field extension K'/K and a morphism $\mathrm{Spec}(K') \to S'$ such that the following solid diagram commutes



As the square is cartesian we get the two dotted arrows a', b' making the diagram commute. Since $X' \to S'$ is universally injective we get $a' = b'$, by Morphisms, Lemma 01S4. Clearly this forces $a = b$ (by the discussion in Schemes, Section 01J5). Therefore Lemma 02KP applies and we win.

An alternative proof would be to use the characterization of a universally injective morphism as one whose diagonal is surjective, see Morphisms, Lemma 01S4. The lemma then follows from the fact that the property of being surjective is fpqc local on the base, see Lemma 02KV. (Hint: use that the base change of the diagonal is the diagonal of the base change.) □